

Main Strengths

The proposal does well in these areas:

Clear problem definition

Strong project organization

Good understanding of blockchain concepts

Proper use of hybrid storage (on-chain + off-chain)

Realistic development timeline

Good use of modern technologies like React, Convex, Besu, and MetaMask

Strong focus on academic integrity and secure communication

Clear explanation of user workflow and system architecture

The project is practical as a prototype and demonstrates strong research and software engineering ability.

Major Weaknesses and Fixes

1. The BB84 System Is Only a Simulation

Problem

The proposal describes the system as "quantum secure," but the implementation is only a software simulation of BB84 using JavaScript.

There is:

no quantum hardware

no photons
no real quantum channel
no actual quantum eavesdropping detection

So the system is not true Quantum Key Distribution (QKD).

Fix

Reword the proposal to describe it as:

"BB84-inspired"
"quantum-inspired"
"educational simulation"

instead of claiming real quantum security.

2. The System Is Not Truly Post-Quantum Secure

Problem

The proposal uses:

MetaMask
ECIES
Ethereum wallets
elliptic curve cryptography

These are vulnerable to future quantum attacks using Shor's Algorithm.

Fix

Explicitly state that:

current wallet authentication is still quantum vulnerable

future versions should use post-quantum algorithms like Kyber or Dilithium

This improves academic honesty and technical accuracy.

3. BB84 Logic Is Conceptually Redundant

Problem

You encrypt the BB84 bit arrays using ECIES before transmission.

That means classical encryption already secures the communication, making BB84 unnecessary.

Fix

Present BB84 as:

an educational demonstration

a conceptual simulation

rather than the actual source of security.

4. Blockchain Justification Needs Improvement

Problem

An examiner may ask:

"Why use blockchain instead of a secure centralized database?"

The proposal does not strongly defend this choice.

Fix

Explain that blockchain provides:

decentralized verification

tamper-proof audit trails

reduced institutional control over records

while also acknowledging that centralized systems are simpler.

5. MetaMask May Be Difficult for Students

Problem

Most students and lecturers are unfamiliar with wallets and private keys.

This creates usability issues.

Fix

Suggest future improvements like:

institutional login integration

automatic wallet generation

simplified onboarding

6. Missing Threat Model

Problem

The proposal lacks a formal cybersecurity threat model.

It does not clearly define:

attackers

insider threats

stolen wallets

malicious validators

phishing attacks

Fix

Add a dedicated "Threat Model" section explaining:

possible attackers

attack methods

security assumptions

7. Metadata Is Still Exposed

Problem

Even though messages are encrypted, attackers can still see:

sender identities

receiver identities

timestamps

communication patterns

Fix

Acknowledge metadata leakage and mention future privacy improvements like:

stealth addresses

zero-knowledge proofs

mixnets

8. The Network Is Not Fully Decentralized

Problem

The Besu blockchain is permissioned and controlled by the university.

So the system is only partially decentralized.

Fix

Use more accurate terms like:

"permissioned distributed network"

"partially decentralized infrastructure"

instead of "fully decentralized."

9. No Wallet Recovery Mechanism

Problem

If users lose their MetaMask recovery phrase, they permanently lose access.

Fix

Suggest future recovery methods such as:

institutional recovery systems

social recovery

backup key systems

10. Legal and Compliance Concerns

Problem

Blockchain immutability can conflict with privacy laws like GDPR.

Fix

Clarify that:

only hashes are stored on-chain

personal data remains off-chain

off-chain data can be deleted when necessary

11. Performance Assumptions May Be Unrealistic

Problem

Timing estimates may vary depending on:

internet speed

browser performance

mobile devices

MetaMask delays

Fix

State that the timings are:

"expected under controlled testing conditions."

12. Weak Evaluation Methodology

Problem

The proposal lacks detailed testing metrics.

Fix

Add:

latency testing

encryption benchmarks

usability measurements

scalability testing

13. Smart Contract Design Is Incomplete

Problem

The proposal never explains:

contract structure

functions

events

access controls

Fix

Include a simple smart contract model with:

message hash storage

verification functions

timestamps

14. "Zero Cost" Is Misleading

Problem

Even without gas fees, the blockchain still requires:

servers

maintenance

storage resources

Fix

Clarify that:

transactions are free for users

infrastructure still has operational costs

15. Security Claims Are Overstated

Problem

Terms like:

"unbreakable"

"perfectly secure"

"cannot be decrypted"

are too absolute for academic cybersecurity writing.

Fix

Use more professional terms like:

"quantum-resistant"

"designed to resist"

"computationally secure"

Final Evaluation

The proposal is strong as:

a prototype

a proof-of-concept

an educational blockchain security project

Its biggest weakness is overclaiming real quantum security.

If you:

- reduce exaggerated claims
- clearly explain limitations
- acknowledge the BB84 simulation
- improve the security discussion

then the proposal becomes far more academically credible and technically defensible.

Most Important Corrections

The five most critical improvements are:

1. Stop calling the system fully quantum secure
2. Clearly state BB84 is simulated
3. Admit MetaMask and ECIES are not post-quantum secure
4. Add a proper threat model
5. Remove terms like "unbreakable security"